

Derek Roberts

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EDUCATION

University of Pennsylvania

Master of Science in Engineering in Artificial Intelligence

Aug 2026 - Present

Remote

Boston University

Bachelor of Arts in Computer Science | GPA: 3.67 | Dean's List (4 semesters)

Aug 2020 - May 2024

Boston, MA

PROFESSIONAL EXPERIENCE

Full-Stack Software Engineer

John Hancock

June 2024 – Present

Boston, MA

Cognitive AI Team

- Own end-to-end production deployments for globally distributed virtual assistants across John Hancock and Manulife, ensuring reliability and minimal downtime.
- Led migration of all chatbots from Microsoft Bot Framework to the Microsoft Agents SDK, enabling agentic AI architectures and modernizing the platform.
- Spearheaded adoption of standardized Git workflows, improving release reliability, code review quality, and team-wide developer efficiency.
- Building a new customer-facing chatbot for the Manulife homepage using Azure, C#, and the Microsoft Agents SDK, driving the shift from predictive to agentic AI.

Finance & Tax Team

- Engineered enhancements to the MODCO (Modified Coinsurance) application using TypeScript, SQL, Azure, and C#, supporting critical finance and tax reporting workflows.
- Designed SQL data transformation and aggregation pipelines and Power BI dashboards to surface actionable financial insights for stakeholders.
- Built a Power Automate flow to fully automate MODCO report generation and client distribution, eliminating manual effort and reducing turnaround time.
- Developed a matching algorithm for the Manulife Women in Tech mentorship program, optimizing mentor-mentee pairings across role, skills, location, and preferences.

Software Engineering Intern

Salary.com

May 2023 – December 2023

Boston, MA

- Rebuilt a QA testing pool UI in ReactJS with integrated web APIs, improving test coverage workflows and reducing manual data entry for the QA team.
- Designed and implemented a marketing popup from concept to production, applying UI/UX principles to drive user engagement and conversion.
- Contributed across the full stack, gaining hands-on experience with front-end development, web APIs, and internal IT systems in a production environment.

Software Project Management Intern

BU Spark!

September 2023 – May 2024

Boston, MA

- Managed a portfolio of client software projects, aligning student engineering teams with partner expectations to consistently deliver on scope and quality.
- Guided the team to “Best Project” in both semesters; the “Computerized Mapping of Visual Fields” project was selected for the Neuroscience of the Everyday World Conference 2024.
- Awarded Spark’s “Firestarter” award for outstanding contributions to the organization.

TECHNICAL SKILLS

AI / Cloud: Azure, GCP, Conversational AI, Agentic AI, Microsoft Agents SDK, Microsoft Bot Framework

Back-End: C#, Python, Java, SQL, .NET **Front-End:** TypeScript, JavaScript, HTML, CSS, ReactJS

Tools: Git, VSCode, SSMS, Power BI, Power Automate, JIRA, TeamCity, New Relic, LaTeX, Microsoft Office

CERTIFICATIONS

Introduction to Terraform on Azure | CSSLP Cert Prep: Secure Software Concepts | CSSLP Cert Prep: Secure Software Requirements | CSSLP Cert Prep: Secure Software Design | CSSLP Cert Prep: Secure Software Implementation | Learning the OWASP Top 10 | C# and .NET Development with Visual Studio

RELEVANT COURSEWORK

AI Engineering | AI and Machine Learning | Intro to Algorithms | Data Structures & Object-Oriented Programming | Combinatoric Structures | Computer Systems | Probability in Computing | Software Engineering | Intro to Databases